Available online at www.sciencedirect.com

SciVerse ScienceDirect

Procedia - Social and Behavioral Sciences 46 (2012) 3345 – 3349

Procedia
Social and Behavioral Sciences

WCES 2012

What do players (think they) learn in games?

Selen Turkay *, Sonam Adinolf

Teachers College Columbia University, New York, 10027, USA

Abstract

The goal of this study is to contribute to the growing literature on games and learning. We will discuss results from an exploratory study where we asked gamers' opinions about learning in games. The data was collected as part of a larger survey study on a variety of topics about games (N = 769). Four themes emerged from the data: 1) players learn about and from game mechanics, 2) players learn from game narratives, 3) players learn from each other, and 4) players learn by becoming interested in an in-game topic and expand their knowledge by studying outside resources.

© 2012 Published by Elsevier Ltd. Selection and/or peer review under responsibility of Prof. Dr. Hüseyin Uzunboylu
Open access under CC BY-NC-ND license.
Keywords: Games, informal learning, survey study, player perceptions

1. Introduction

During the last two decades, thanks to the advancements in information communication technologies, digital games have become more popular for school aged children then ever (Pew Research, 2008), and there is a growing body of evidence suggesting that digital games have educational value (see de Freitas, 2008). This has led to excitement about the possible uses of such games for teaching and learning (Gee, 2003; Kafai, 2006). Consequently, the number of research studies in this area has also increased dramatically.

We identified five main research branches in games and learning literature: 1) developing and testing effectiveness of educational (e.g., Dede et al., 2005; Rosenbaum, Klopfer, & Perry, 2007) or serious games (Nilsen et al., 2011) in formal educational settings; 2) using commercial-off-the-shelf (COTS) games in formal educational settings (e.g., Adams, 1998; Squire, 2004); 3) studying games, both theory based custom-built and COTS, as part of informal learning (Goldman, Koepfler & Yocco, 2009); 4) creating pedagogical models that we can draw from videogames (e.g., Becker, 2006; Gee, 2003); 5) having students design and develop games for learning in formal educational settings (Kafai, 2006).

In this exploratory study, we are interested in incidental learning with digital games in the context of informal learning. In particular, we will take a close look at players' perception of learning with digital games (e.g. what people think that they learn when they play games in their leisure time). We think that more evidence is needed in the area of games as informal learning tools. Therefore, our research questions were:

- Do players think they learn when they play games? If yes, what do they learn and how do they learn?
- Is there any age and gender difference in players 'attitude towards games for learning?

After a brief overview of related studies, we will present results of the survey study followed by a discussion where we will discuss findings in relation to transfer and provide implications for game design.

* Selen Turkay. Tel.: +0-212-678-3807
E-mail address: st2282@columbia.edu

1877-0428 © 2012 Published by Elsevier Ltd. Selection and/or peer review under responsibility of Prof. Dr. Hüseyin Uzunboylu
Open access under CC BY-NC-ND license. doi:10.1016/j.sbspro.2012.06.064

2. Background

It is quite difficult to separate the process of learning from the practice of education (Sefton-Green, 2004). As a consequence, work in the field of games and learning has mainly focused on developing educational games for teaching school subjects; literature mainly consist of studies which were conducted either in school or after-school programs with custom or COTS games which were aimed to be used as tools for producing authentic participation.

Previous studies using COTS games in classroom settings have been largely successful. Adams (1998) used *SimCity 2000* to give undergraduates a greater appreciation of the problems and responsibilities of urban planners. Squire (2004) investigated *Civilization III* as a means to improve students' critical thinking about history. Although these and similar studies found positive learning outcomes with COTS games, researchers also emphasized the challenge of integrating those games into school curriculum (Foster & Mishra, 2009; Squire, 2005). As a result, many researchers are testing the effectiveness of custom-built games in various range of studies. Examples of this include teaching computational strategy with a racing game (Holbert & Wilensky, 2011) and a game called Urban Science putting students in the role of urban planners (Bagley & Shaffer, 2009). Other areas of study that researchers have used custom games in a structured setting are teaching probability (Lee, Lee, & Lau, 2006), concept of Relativity (Carr & Bossomaier, 2011), and coping with bullying in schools (Aylett et al., 2006).

Considering that the majority of human life, and therefore learning, takes place outside of formal educational institutions, the question remains how much of this learning might be gained from playing games. Because of the methodological problems of researching learning in informal settings, there are few experiemental studies in the area of learning with games in informal settings. Research on informal learning with games can be also examined in two parts. The first is the research with games that were designed for informal learning with educational theories in mind. The other is the research with COTS games.

Multiple theory based educational games (e.g., *DinoQuest Online* (DQO), *Food Force*, *Whyville*, *WolfQuest*, *ImmuneAttack*) have been developed with consideration of making them motivational so that they can be played both in and out of school. *WolfQuest*, an educational massively multiplayer online game (MMO) about wolves, is an example to such effort. A summative evaluation report on *WolfQuest* presents evidence on its effectiveness in achieving its educational goals. Among those goals was motivating players to research about wolves in outside resources. They also found that the more players played the game the more they showed this kind of behaviour (Goldman, Koepfler & Yocco, 2009). Similar affordances were identified for DQO which was designed for informal science education in the domains of life science and paleontology for K-6th grade students (Scacchi, 2010).

One of the aspects of good game design is replayability. Unfortunately, custom-built educational games usually fail in retaining players when competing with commercial titles. For example, Barendregt & Bekker (2011) found that even though students liked *Hello You!* game more than other activities for learning English at school, they did not choose to play the game in their leisure time, mostly because it was not engaging enough for them.

Close inspection of the studies with COTS games in informal learning emphasizes the importance of social aspect. Researchers used ethographic research methods to look for evidence of various kinds of learning in MMOs (see Steinkuehler, 2004). Gaming forums is another place where informal learning occurs (Turkay et al., 2010). Steinkuhler and Duncan (2008) analysed 1,984 forum posts made by 1087 unique game characters on *World of Warcraft*'s official forums. They showed evidence on MMO players' scientific habits of mind including model-based learning, use of evidence and social construction of knowledge. It was proposed that in order for useful learning to take place, peer interaction and incidental learning must be part of educational process. This resembles the learning ecology around video games where peer learning dominates the informal lerning practices and social interaction mediates the game experience (Jessen, 2001, as cited in Egenfeldt-Nielsen, 2006).

Incidental learning occurs through one's experiences, including mistakes, successes, and interactions with others (Marsick & Watkins, 1990). We believe that incidental learning might be one of the many ways players learn in video games. While players develop skills in expert thinking and complex communication, they also get ready to meet the needs of the 21st century. These aspects have been also given as strength when emphasizing pedagogical strengths of video games as learning tools (Gee, 2003; Prensky, 2001).

We believe that literature of informal learning with games can benefit from further studies which employ different methodologies. In this study, instead of an ethnographic or an experimental study, we conducted a survey study with players. The next section details our methodology.

3. Methodology, Participants and Data Analysis

A mixed method survey was used to collect data from online forums through snowball sampling. These forums were public and private gaming forums. A total of 769 players filled out the survey. The survey collected demographic data as well as participants' weekly gaming hours and their favorite game genres. We asked participants three questions regarding games and learning: 1) Have you learned anything new in games?; 2) Can you tell us one of the things that you have learned recently?; 3) Are games good for learning? Why/why not?

Data was analyzed using the quantitative data analysis software SPSS 18.0, and answers of the open-ended questions were analyzed thematically by using Nvivo 9 qualitative analysis software.

4. Findings

When we asked participants whether they think that games are good for learning, 91% (n = 699) reported that games are in fact good for learning and they have learned various kinds of information, knowledge and skills as a result of playing games. The rest fear when games become educational, they would aim to teach players the content not in a fun way. This goes against the perception that players will avoid educational games because they're *educational*. The more obvious reason for players to avoid many educational games is that they're not *fun*. This supports Squire (2011, p.87) where he states that people don't think learning in games is weird or fun. Similarly, a previous survey study with young adults and their mobile phone use found that they would be interested in learning math, literacy or a foreign language with mobile games if they were appealing, relevant and fun (Attwell & Savill-Smith, 2004). Interestingly, players think that the more difficult games are, the better they are for learning because they require thinking and reflection.

We found a significant correlation between how much participants think games are definitely good for learning and who reported that they have learned things from games ($p < 0.001$; $r = 0.424$). There was no statistically significant gender difference in this result. There was a statistically significant difference among different age groups. Tukey's Post hoc test showed that significantly more players who are 18 and below reported that they have learned various things from games compared to players between 19 and 30 ($p < 0.05$).

Out of 699 participants who said games are good for learning and they learned from games, 478 gave examples on what they learned in games. Analysis of these examples revealed how and what players learn from games.

4.1. How do players learn?

4.1.1. Players learn from game mechanics

This seems quite obvious, as players need to know about game mechanics to play the game. While players learn semiotic domains of games (Gee, 2003), they can also learn knowledge such as physic laws and develop skills such as mathematical thinking, and faster reflexes. For example, in Sid Meyer's *Pirates*, a player is given a wind meter which could teach about force vectors. Firing cannons from the side of ships can teach about perpendicular lines. What different types of ships and nationalities were present also communicates to the player in the game.

"I've improved my mathematical thinking… Comparing statistics of my equipment, trying to figure out how to best spend my points on characters that gain levels and are customizable... In war games, there's financial planning, and numbers play a big part in your strategy…" – Male, North America.

4.1.2. Players learn from game narratives

Although not present in all games, narrative is a critical element in many modern video games. While the narrative creates an immersive and engaging environment, it can also facilitate learning. Participants reported learning new vocabulary and improved reading skills from written dialogue or mission text. They learned about historical events from the setting of the game, and details in the story. Assassins' Creed 2, which takes place in Renaissance Italy, was given as an example by participants who reported learning from the story of the game.

"Red Dead Redemption, for example, is a game that is set in the early 1900's. Since pretty much EVERY gamer wasn't even alive back then, it can teach you interesting things about the Old West like that automobiles were only just beginning to be mass-produced and that most people got around by horseback." – Male, North America.

4.1.3. Players learn from each other

Thirty one percent of the participants reported learning that mainly originates from the social aspect of games. As we expected, players with their favorite games of MMORPGs mentioned social learning significantly more than other players ($p<0.001$; $t = 9.407$). Participants reported a wide range of skills and knowledge they learned from games. For example, leadership, communication, teamwork skills, politics and cultures, internet resources for games, recipes and computer hardware are among those. Some players even reported getting recipes for certain dishes or even getting violin lessons over voice chat. Supporting Steinkuehler & Duncan (2008)'s finding, participants reported gaming forums as the main place where they construct their knowledge about game mechanics.

"One day in kinship chat, people were discussing college options, and I learned a lot about the flexibility of a lot of schools in terms of class times and etc." – Female, North America.

"I had a long conversation with the best rogue on the server… One time we talked for about 3 hours about different rogue mechanics and I learned a lot but I definitely think so did he…" – Male, North America.

4.1.4. Players learn through tangential learning

The last category is tangential learning (Floyd & Portnow, 2008). Players learn by becoming interested in an in-game topic, and expand their knowledge on this topic by studying outside resources without obligatory reinforcement. Similar to self-directed learning, players reported educating themselves when introduced to topics in a context that they are already engaged in. For example, players may become interested in Greek mythology after playing *God of War* and may learn how to create machinima when inspired by in-game videos.

"Valkyrie Profile made me interested in Mythology" – Male, Asia.

4.2. What do players learn?

Data revealed three major categories of skills learned in games: 1) social skills such as team-work and negotiation; 2) motor and spatial skills such as getting better at reflexes, navigation and eye-hand coordination; 3) cognitive skills such as problem solving, estimation, languages, and strategy development. Players also reported gaining various personal traits such as patience, trusting others and persistence.

Players learn both declarative (e.g., historical facts) and procedural knowledge (e.g., how to negotiate) when they play games or in the affinity space of a game (Gee, 2003). Among the knowledge participants reported are mathematics and statistics (37%), science (3%), improving vocabulary and foreign language skills (12%) and history (9%). Thirty two percent reported learning about the game, such as strategies and the game plot. The rest include variety of areas such as geography and daily life related things such as politics and cultural values.

5. Conclusion and Discussions

As video games in education are gaining more attention, it becomes critical to understand how games can impact students learning out of school and help them develop life-long learning skills. Our results provide promising evidence that players not only learn in games but they also extend their knowledge outside of the game when they get interested in an in-game topic. Games in general teach goal setting, perseverance, resource management, and encourage persistence. If educational game designers aim for transfer from games, they need to design games that are challenging, and that encourage post-game reflections and tangential learning.

We often hear that educators or game designers say: "do not tell students this is an educational game. They will not play it." This exploratory study shows evidence to the contrary. We believe that as long as games are engaging and fun, students will play them and if the opportunities arise, they will use the knowledge and skills they learn from games. The following quote from a school age participant is representative: "Games could potentially be good for learning, if you made the game fun. Since the interactivity of a game, along with how fun it is, can really make a player interested." This may mean that we don't need stealth learning in games, we need good design.

If people are motivated to do something, it is likely that they will be willing to spend time on it and they may learn from it. That is why people spend hours on hobbies like playing games, and game designers are responsible from what kind of learning can be gained from games. We believe that more games should be designed with the goal of promoting incidental and tangential learning. Gee (2008) identified five conditions for an experience in a

game that needs to be useful for learning: specific goals, immediate feedback, application of previous experience, learning from peers and experts, and interpretation. We want to add a few conditions which came up in our study: add complexity and challenge (good level design does it), make players curious or surprise them, make the learning part of the game mechanics, and expose players to new themes and concepts. These can make a useful checklist when designing a game experience that will be a part of players learning ecology.

References

Adams, P.C. (1998). Teaching and learning with SimCity 2000. *Journal of Geography, 97*(2), 47–55.

Atwell, J. Savill-Smith, C. (Eds.). (2004). *Learning with mobile devices: Research and Development.* Learning and Skills Development Agency.

Aylett, R. S., Figueiredo, R., Louchart, S., Dias, J., & Paiva, A. (2006). Making It Up As You Go Along - Improvising Stories for Pedagogical Purposes. In *Proceedings of the 6th International Conference for Intelligent Virtual Agents (LNAI 4133),* 307-315. SpringerVerlag.

Bagley, E., & Shaffer, D.W. (2009). When people get in the way: Promoting civic thinking through epistemic gameplay. *International Journal of Gaming and Computer-mediated Simulations, 1,* 36–52.

Barendregt, W. & T. M. Bekker (2011). The influence of the level of free-choice learning activities on the use of an educational computer game. *Computers & Education, 56*(1), 80-90.

Becker, K. (2006). Pedagogy in Commercial Video Games. In D. Gibson, C. Aldrich & M. Prensky (Eds.), *Games and Simulations in Online Learning: Research and Development Frameworks* (pp 21-48). Hershey, PA: Idea Group Inc.

de Freitas, S. and Griffiths, M. (2008). The convergence of gaming practices with other media forms: what potential for learning? A review of literature. *Learning, Media and Technology, 33* (1),11-20.

Carr, D. & Bossomaier, T. (2011). Relativity in a rock field: A study of physics learning with a computer game. *Australian Journal of Educational Technology, 27*(6), 1042-1067.

Dede, C., Ketelhut, D. J., Clarke, J., Nelson, B., & Bowman, C. (2005). *Students' motivation and learning of science in a multi-user virtual environment.* Paper presented at the American Educational Research Association Conference, Montreal, Canada.

Egenfeldt-Nielsen, S. (2006). Overview of research on the educational use of video games. *Nordic Journal of Digital Literacy, 3*(1).184-312.

Floyd, D. & Portnow, J. (2008) Video Games and Learning [Online video]. Available from http://www.youtube.com/watch?v=rN0qRKjfX3s

Foster, A. N., Mishra, P. (2009). Games, claims, genres & learning. In R. E. Ferdig (Ed.), *Handbook of research on effective electronic gaming in education* (pp 33-50). IGI Global.

Gee, J. P. (2003). *What video games have to teach us about learning and literacy.* New York: Palgrave/St. Martin's.

Gee, J.P. (2008). Learning and games. In Salen K (Eds.) *The ecology of games: connecting youth, games, and learning* (pp 21–40). MIT Press, Cambridge, MA.

Goldman, K.H., Koepfler, J. & Yocco, V. (2009). WolfQuest Summative Evaluation: Full summative report. Institute for Learning Innovation. Retrieved from http://informalscience.org/reports/0000/0206/WQ_Full_Summative_Report.pdf

Holbert, N. R. & Wilensky, U. (2011). *Racing games for exploring kinematics: A computational thinking approach.* Paper presented at the American Educational Research Association Conference, New Orleans, LA.

Kafai, Y. B. (2006). Playing and making games for learning: Instructionist and constructionist perspectives for game studies. *Games and Culture: A Journal of Interactive Media, 1*(1), 36-40.

Lee, J. H. M., Lee, F. L., & Lau, T. S. (2006). Folklore-based learning on the web—pedagogy, case study, and evaluation. *Journal of Educational Computing Research, 34*(1), 1-27.

Marsick, V.J. & Watkins, K. (1990). *Informal and incidental learning in the workplace.* New York, Routkedge.

Nilsen, E., LeDonne, R., Klemperer, E., & Olund, S. (2011). *Conflict resolution with a serious game for peace*. Paper presented at Games, Learning, and Society 7.0 Conference. Madison, WI.

Pew Research (2008). Teens, video games and civics: Teens' gaming experiences are diverse and include significant social interaction and civic engagement. Retrieved from http://www.pewinternet.org/PPF/r/263/report_display.asp.

Prensky, M. (2001). *Digital game-based learning*. New York, NY: McGraw-Hill.

Rosenbaum E., Klopfer E., Perry J. (2007). On location learning: Authentic applied science with networked augmented realities. *Journal of Science Education and Technology, 16*(1), 31–45.

Scacchi, W. (2010). Game–based virtual worlds as decentralized virtual activity systems. In: W.S. Bainbridge (Eds). *Online worlds: Convergence of the real and the virtual* (pp. 223–234). New York: Springer–Verlag.

Sefton-Green, J.(2004). *Literature Review in Informal Learning with Technology Outside School*. Bristol, England: Futurelab.

Steinkuehler, C.A. (2004) Learning in Massively Multiplayer Online Games.In *Proceedings of the 6th international conference on Learning Sciences*, (pp 521-528). Santa Monica, CA.

Steinkuehler, C. A. and Duncan, S. C. (2008). Scientific habits of mind in virtual worlds. *Journal of Science Education and Technology, 17* (6), 530-543.

Squire, K. (2004). Replaying history. Unpublished dissertation, Indiana University, Bloomington.

Squire, K. (2005). Changing the game: What happens when video games enter the classroom? *Innovate, 1* (6). Retrieved from http://www.innovateonline.info/index.php?view=article&id=82

Squire, K. (2011). *Video Games and Learning: Teaching and Participatory Culture in the Digital Age.* New York, NY: Teachers College Press.

Turkay, S., Kinzer, C., Hoffman, D., Gunbas, N. & Nagle, C. (2010). A Snapshot on Youths' Activities on Online Gaming Forums: Internet and Informal Learning. In *Proceedings of World Conference on Educational Multimedia, Hypermedia and Telecommunications 2010* (pp. 3987-3992). Chesapeake, VA: AACE.